

Final Report Summary for Internal Distribution

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Project Title	An Integrated Discovery Workflow for an E.coli Whole-Cell Model
OT Award #	HR0011-24-3-0372
Award Start Date	27SEP2024
Award End Date	27SEP2025

1 Executive Summary

Motivated by the goal of computational causal discovery for cell biology, this project advanced the feasibility of simulation-based inference for whole-cell models (WCMs), by identifying the primary numerical and computational obstacles faced by a state-of-the-art WCM of *E. coli* [1], as implemented in the multi-paradigm compositional simulation engine Vivarium [2].

While the original proposal expected to match the goals of the DUF ARC with a large-scale computational workflow that derived hypothesized gene function annotations from purely synthetic WCM simulations, an in-depth statistical analysis led to the conclusion that, within the award's timeline and compute budget, such an inverse problem formulation is not currently tractable without substantial refactoring of the WCM's numerical (Sec. 4.1.1) and software structure (Sec. 4.2.1). Accordingly, and in alignment with ARC's risk-reduction intent, we reoriented the remaining effort toward a systematic investigation and prototyping effort, for rebuilding a dynamic WCM with well-behaved inverse problems that can eventually support rigorous functional discovery methods based on dynamic causality (Sec. 3).

At the infrastructure level, a methodology was proposed for transforming the heuristic discrete-time simulation engine of the current mechanistic WCM into a continuous-time stochastic dynamical system simulator in the class of piecewise-deterministic Markov processes (PDMPs), a.k.a. jump-ODEs (Sec. 4.1.2). While maintaining the basic abstraction of discrete-event simulation that is at the core of Vivarium, enforcing this stochastic process structure will enable more efficient numerical simulation algorithms that intrinsically produce causally consistent Bayesian likelihoods. A refactoring of the current algebraic representation of topological composition will furthermore enable compiled execution in tandem with high-performance numerical integrator libraries (Sec. 4.2.2). At the application level of the WCM, the transformation starts with the most substantial process reformulation, namely of the core

metabolism from a tFBA model [3] into an ODE model, and will incrementally expand outwards in future projects (Sec. 4.1.2).

While large-scale parameter inference was not yet performed, these efforts established a rigorous numerical foundation for linking mechanistic hybrid cell simulations to statistical modeling, isolated the computational bottlenecks limiting current approaches, and identified a modular architecture that unifies simulation, data standards, and rigorous inference for next-generation WCMs. Our recommendations for future research correspond to several distinct multidisciplinary research projects, and include an incremental path toward a full WCM reformulation, leveraging the modularity of the current WCM. In closing, we discuss estimated resource requirements for future work (Sec. 5.3).

2 Background

2.1 Established Forward Models for Cell Simulation

Kinetic models of metabolism. Kinetic models describe the temporal dynamics of metabolite concentrations and enzyme activities through systems of ordinary differential equations (ODEs). Each reaction is parameterized by experimentally measured or estimated rate constants, enabling mechanistic simulations of transient responses and regulatory effects. The scope of these models, however, is limited by incomplete kinetic characterization, and most rate constants (e.g., k_{cat} and K_m for Michaelis-Menten kinetics) remain unknown or context-dependent. Nonetheless, highly curated models, such as [4] for *E. coli* central metabolism, show that even partial knowledge of kinetic parameters and a modest amount of data, when combined with strongly informed priors about the functional forms of kinetic equations, can reproduce coordinated glucose uptake, energy metabolism, and growth regulation.

Constraint-based models of metabolism. Constraint-based approaches, including flux balance analysis (FBA) and its dynamic extensions (dFBA, tFBA, CAFBA), treat metabolism as an optimization problem constrained by stoichiometry, mass balance, and thermodynamics. These methods predict flux distributions and growth rates without detailed kinetics and scale efficiently to genome-wide reconstructions of *E. coli*. Time-linked FBA (tFBA) [3], used in the current Vivarium *E. coli* model, introduces time-dependence by solving sequential optimization problems, allowing simulation of nutrient depletion and growth. However, this formulation introduces discontinuities (e.g., a jump between facets of the flux balance polytope within a discrete-time step) and control terms (e.g., target concentrations, target fluxes, target doubling time) that are difficult to reconcile with continuous-time causal semantics and that do not directly induce a likelihood suitable for Bayesian inference, which complicates statistically grounded inverse problems.

WCMs. Whole-cell models integrate metabolism, gene expression, DNA replication, and molecular assembly into a unified computational framework that simulates an entire cell life cycle. The *E. coli* WCM, implemented in Vivarium [2] and detailed by [1], organizes biological processes as modular algorithmic processes interacting through shared states, implemented in Vivarium as stores. This modular architecture enables distributed execution, composability, and reuse, supporting simulation of over 4,000 gene products and tens of thousands of molecular species. The WCM represents the most comprehensive mechanistic integration of biological

knowledge for any organism to date, yet it is best characterized as a hybrid algorithmic simulator, rather than as a numerical integrator of a mathematically defined dynamical system. It mixes ODE-based, stochastic, and optimization-based components, without a single underlying probabilistic or differential formulation. Consequently, while forward simulations capture known phenotypes across approximately a dozen environmental conditions, they cannot yet quantify uncertainty over internal states, infer missing parameters, or discover unknown functions.

2.2 Established Inference Methods for Cell Simulation

Kinetic models of metabolism. Parameter inference for kinetic models typically relies on global optimization or Bayesian sampling methods such as Markov Chain Monte Carlo (MCMC), which aim to fit model trajectories to time-course metabolite or flux data [5], effectively assuming that neither other cell processes nor bulk-level heterogeneity exist. Furthermore, while these methods provide interpretable parameter posteriors and enable uncertainty quantification within the scope of such simplifying assumptions, they are computationally expensive and difficult to scale with model dimensionality. For genome-scale systems or even moderately large metabolic networks, Bayesian inference is thus impractical without highly customized methods, which leverage algebraic decompositions of the probabilistic model structure and iteratively construct variational approximations.

Constraint-based models of metabolism. Constraint-based models, including FBA and its dynamic extensions, take a complementary approach. Rather than estimating kinetic parameters, these models are parameterized primarily by stoichiometric and thermodynamic constraints, and an experimentally derived steady-state objective function such as growth or biomass production. In this context, inference usually refers to the estimation of internal flux distributions that are consistent with observed exchange fluxes, steady-state assumptions and evolutionary optimality assumptions. This is typically achieved through linear or quadratic programming formulations, often coupled with heuristic hyperparameter fitting. While computationally efficient, constraint-based methods are designed primarily to support decision-making for bulk-level metabolic engineering, and typically lack a probabilistic interpretation that would capture the dynamic variability or stochastic effects observed in living systems. A notable recent development is an MCMC method applied to the bulk-level flux space of a genome-scale model [6], using as its likelihood Elementary Metabolic Unit (EMU) simulations with an ad-hoc Gaussian noise model.

WCMS. Whole-cell models integrate multiple biological subsystems, such as metabolism, gene expression, DNA replication, and molecular assembly, each with their own parameters and data sources. Most WCM parameters are drawn directly from highly curated experimental literature, or are optimized at the subsystem level based on local steady-state assumptions, while a small number of key coupling parameters (e.g., weights of homeostatic and kinetic tFBA objectives) are optimized globally to reproduce viable growth and agreement with aggregate datasets at the WCM level. Additional calibration is performed by combining global sensitivity analysis with full factorial design, in order to identify localised causes for robustly mismatched model predictions, thus motivating targeted new measurements [1]. This methodology relies heavily on expert judgement at every step, and while it yields models that are computationally predictive within their scope of calibration, they are difficult to characterize in their statistical generalizability, as they form a complex multi-layer network of hard and soft constraints, involving instantaneous and intertemporal dependencies, as well as environment-specific re-calibrations (ParCa – “parameter calculator”).

3 Objective and Approach

Causal discovery problem. In the context of functional discovery, the *manual design* both of laboratory experiments and of computational experiments relies on hypothesized correlations between localized interventions and targeted measurements, and the conclusiveness of such studies hinges on whether the hypothesized correlations are strong enough to *experimentally dominate* the scale of, and the uncertainty over, other coexisting processes and effects. This scientific methodology has also been essential for the traditional development and calibration of WCMs. In contrast, the appropriate *statistical formulation* of a more heavily data-driven workflow for functional discovery would be *simulation-based causal discovery*, lifting WCMs into statistical models and shifting some of the burden of confounder control to numerical methods. Our original proposal thus envisioned incorporating an established WCM as a black box into a probabilistic programming language (PPL) with causal modelling capabilities, e.g., ChiRho [7], and to quantify the effects of many candidate annotations in a setting of purely synthetic data.

Numerical challenges. Subsequently, it became apparent that statistically robust methods for causal discovery, in order to *quantify the evidence* for each alternative causal mechanism under intrinsic parameter uncertainty, necessarily have to marginalize over relevant parameter subspaces and latent dynamics [8-11]. For WCMs, this marginalization is not only very high-dimensional, but requires, particularly in the case of causal discovery for biochemical functions with high kinetic uncertainty, the approximate solution of many Bayesian parameter inference subproblems in an inner loop [12]. This need is exacerbated by literature demonstrating that kinetic models of metabolism can exhibit strong self-regulation, including some capabilities previously attributed to genetic regulation [4], which suggests that the functional form of the current tFBA model of metabolism may act as a major confounder in causal discovery attempts (see DOE, MS 5 Technical Report). However, a nested inference problem of this kind is computationally expensive already for very simple models (see, e.g., working examples in ChiRho [7]), and is currently infeasible for WCMs, in which parameter inference alone is notoriously difficult [13].

In contrast, previous work on this WCM approached parametrization by heavily relying on curated databases of function annotations and kinetic measurements, developed bespoke sensitivity analysis and calibration methods only for a small number of critical parameters while aiming only at point estimates that support viability constraints, and relied for validation on omics data with very low sample size per environment condition. In other words, the original project scope depended internally on a type of uncertainty quantification that had not been addressed previously for this WCM, and that required much more substantial investment into Bayesian inference methodology and infrastructure than anticipated.

Project rescoping. Thus, we faced the following alternatives for the remainder of the limited project timeline (see also MS 5 Technical Report):

1. Changing the project scope to an application or extension of existing methods for gene function estimation, e.g., based on deep embeddings of static data in enzyme databases [14], or based on combinatorial optimization over stationary FBA models [15], with a potential generalization to parameter ensembles [16]. Such an approach would have remained closer to the nominal goal of DUF ARC, but would have meant entirely

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dropping the core premise of our original proposal, namely the pursuit of inference in a dynamic WCM.

2. Reducing the project scope to a Bayesian parameter inference problem over a very restricted set of parameters. The latter problem may then be solved by embedding the WCM as a black box into an existing PPL, using a simple summary statistic as a pseudo-likelihood, and running an approximate Bayesian computation (ABC) algorithm over WCM simulation samples. Such an approach would mirror the strategy employed in the current WCM to optimize key hyperparameters [1; SM Sec. 1.2.3]. But it would not lead to practical progress towards high-dimensional inference from omics data to WCM parameters, which is a necessity for robust causal discovery workflows.
3. Reorienting the project towards a joint investigation of the numerical structure and software implementation of the existing WCM, identifying key bottlenecks towards the feasibility of inverse problems, and prototyping a narrower forward and inverse problem formulation that can inform future efforts towards WCM improvements and causal discovery workflows.

At the time of the DOE, we selected the third alternative, and narrowed our focus to the core metabolism component, whose dense and fast coupling with all other cell processes makes it the most significant confounder for any parameter inference or causal discovery task. A sequence of prototyping efforts then led to a substantial clarification of the relevant literature, of the numerical problem structure, and of various implementation bottlenecks, which we describe below. Our model integration and parameter inference efforts remained at the prototyping stage, and are intended to inform the design of a *new and structurally distinct* architecture for WCMs, rather than producing improved parameters for the existing WCM.

4 Scientific and Technical Results

4.1 Mathematical Forward Model Structure

4.1.1 Key Identified Challenges

A “mechanistic simulation” is not, in its current form, an explicitly specified dynamical-system model suitable for likelihood-based inference. According to the WCM authors [1; SM Sec. 1.1.1], *“Running our large-scale cell model may be thought of as similar (but not strictly equivalent) to numerical integration of a system of ordinary differential equations, where the cellular states are analogous to the ODE state variables and the cellular processes are analogous to the differential equations. The most significant difference from ODE numerical integration is that shared resources stored in cellular states must be partitioned to each cellular process in order to ensure mass conservation (described in more detail below).”* More precisely, during each fixed-length time step, the WCM combines:

- a modified tFBA step (solving a linear program over a joint space of concentrations and pseudo-fluxes, where the coordinates of the problem encoding depend not only on the time step size, but also on the current actual and “target” values of metabolite concentrations and fluxes),

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- a number of stochastic inter-process couplings (as discrete-time rather than continuous-time jumps, with a parametrization dependent on the ratio of the time step size to the teleonomic “doubling time” parameter),
- and a globally acting discrete-time jump process for resource reallocation, according to “demands” and “priorities” based in a teleonomic / policy-driven rather than derived from a single mechanistic sub-model (serving a role roughly analogous to the corrector step of a predictor-corrector method).

This overall algorithmic dependency structure is not only very difficult to characterize as a mathematical dynamical system model, but has strong and nonlocal numerical artefacts that will impede any global parameter inference effort. At the same time, there are very good reasons for the historical success of such a constraint-based model development approach (see SOP Sec. 9).

Dependence on environmental conditions is lacking in regularity. The current WCM accepts a small set of named environment conditions, each of which is used to separately fit many internal parameters. While dynamically changing between these discrete environments is supported, there are no explicit regularity assumptions for this dependence, practically prohibiting inference across different values of this “global parameter”.

Population dynamics are lacking from current models. Contemporary practice for WCMs in the tradition of metabolic engineering largely ignores the mathematical and statistical distinctions between single-cell and bulk behaviors, both in simulation and in parameter estimation. A rigorous approach should revisit the mathematical biology literature on *physiologically structured population models* [17,18], which further reinforces the need for mathematically well-defined dynamical system models at the single-cell level.

Observation likelihoods are lacking in precision. With notable exceptions [19], measurement likelihoods for omics data of microbial cell cultures are underdeveloped in the existing literature, which is a major bottleneck for statistical inference efforts. Increased independent sample sizes and noise calibration information in published data would be very helpful for estimating likelihoods via empirical Bayes methods. On the one hand, this means that current WCM parameter estimates should be treated effectively as “empirical priors”, i.e., only as representative values for very broad “plausible regions” in parameter space. On the other hand, this calls for focused interdisciplinary collaboration on statistical *measurement models* between statisticians, computational modelers, experimental microbial biologists, and operators of measurement instruments.

Causal modelling frameworks are lacking in structural complexity. It is only in the past decade that the statistical literature on causal modelling has expanded its focus from static to dynamic models, and the latter are currently mostly restricted to structured ODEs [10,11,20]. The next generation of WCMs should aim at a mathematical dynamical system structure that is as close as possible to this literature, to eventually support theoretically justified causal discovery and causal inference methods.

4.1.2 Developed Methodology

Standard stochastic process model class. Upon careful inspection of the modelling assumptions made throughout the current WCM, we identified piecewise-deterministic Markov processes (PDMPs), a.k.a. jump-ODEs, as the most suitable mathematical model class for a systematic reformulation (see SOP Sec. 9). Crucially, PDMPs are backed by unbiased continuous-time stochastic integration methods, they constitute *process likelihoods* by definition (leaving *measurement likelihoods* as a separate issue), and they enforce a clear separation between

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causal and teleonomic parameters. In other words, rather than retrofitting synthetic process likelihoods to the current WCM, we conclude that a scientifically better and overall practically easier goal is the transformation into a PDMP model to which the current WCM relates as an inconsistent discrete-time approximation (see SOP Sec. 9). The main obstacle is the current tFBA model of metabolism [3], for which we were only able to begin a reformulation effort, combining the aforementioned, smaller but highly curated kinetic ODE model [4] with an outer layer of an optimal control ansatz similar to DFBA-LQR [21]. More broadly, and far beyond this WCM of the *E. coli* organism, we believe that PDMPs are an appropriate mathematical abstraction and numerical problem format for biochemical models whose dynamics combine deterministic mass-action kinetics, stochastic monomer and polymer kinetics and piecewise-deterministic cascade control mechanisms, while operating over combinatorially represented state spaces that support local topological changes as discrete events in continuous time.

Incremental model transformation via context-conditional training. Our work towards replacing the dFBA metabolism model led to an essential methodological insight for how to adopt and adapt new model components, wherever they may lie on the spectrum from mechanistic to surrogate. On the one hand, we observed that the development of most of the published mechanistic cell simulation models was based directly on experimental measurements, typically of *in-vitro* kinetics or of bulk omics, which always involves substantial simplifying assumptions. On the other hand, a recent trend of surrogate models uses selective outputs from mechanistic WCMs for training regression models, as low-cost predictors for some *cell-level functional properties* of interest [22]. In contrast to both of these approaches, we note that simulation data recorded at *complete submodel interfaces* (i.e., Markov blankets) within a curated WCM can provide state-of-the-art approximations of the rich correlation structures of *in-vivo* subsystem environments. Beyond its current diagnostic use, such data should be highly valuable for in-context parametrization and validation of new subprocess models, prior to their *embedding within* full WCM simulations. We identified two principal causes for why this potential seems to be currently underutilized: 1) The need to define explicit mathematical and statistical models for *internal interfaces*, and 2) the nontrivial effort to extract and transform simulation data into a suitable numerical format. This overhead can be reduced substantially by simulation frameworks that support the specification of subcomponent interfaces in a unified mathematical format, an approach that Vivarium [2] already takes to the extent of attribute names and data types.

4.2 Software Infrastructure

4.2.1 Key Identified Challenges

Submodel definitions and simulation modules are currently inseparable. Although the Vivarium framework modularizes the *E. coli* WCM into separate processes such as metabolism and transcription, these processes still internally depend on bespoke Python code developed over more than a decade in Markus Covert's Lab [1]. While subprocesses can now be individually executed and replaced, their mathematical model definition cannot be extracted from a process in general, or analyzed or rerun with alternative solvers. As a result, the model/simulation relationship remains inseparable, limiting performance optimization, portability, and integration with modern inference frameworks.

Substantial computational overheads in the current WCM implementation. As a result of long-standing continual development, and notwithstanding speedups achieved by the simulation

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engine Vivarium, the current WCM exhibits orders of magnitude of overhead, relative to competitive numerical solvers that could substitute individual model components, primarily as a tradeoff for modularity and extensibility. A primary cause is that the current design is optimized primarily for modularity, resulting in many avoidable data marshalling operations at each time step, in addition to the essential numerical computations. Because simulation cost is a much more stringent bottleneck for statistical inference than for explorative analysis or for manually guided calibration, future work must prioritize model consolidation into high-performance numerical computing software, orthogonally to parallel computing concerns.

4.2.2 Developed Methodology

Inference-aware simulation platform. Revisiting our original research proposal from a gray box perspective, we find that the requirements for lifting a general-purpose discrete-event simulation engine into a PPL are coextensive with the capacity, as described above, for capturing and processing numerical objects, e.g., likelihood terms, at internal interfaces during simulation. In more technical terms, a programming language theoretic analysis of the Vivarium framework revealed that, while currently not formalized, it may be understood as a domain-specific algebraic effect system [23] for discrete-event simulation. Since algebraic effects are one established strategy for implementing PPLs, as again exemplified by ChiRho [7], this insight opened a clear path for extending the algebraic effects supported in Vivarium towards a PPL. To test this idea, we implemented a prototype for how to add multiple layers of extended algebraic effects on top of a library that was itself a prototype for the next version of Vivarium. However, the main practical difficulties, towards achieving an implementation suitable for simulation-based inference in WCMs, depend on:

- the elaboration of concrete algebraic effects appropriate for the specific statistical model formulation in question, i.e., explicit definitions of the currently implicit mathematical interface types and currently missing likelihood terms between all model components,
- and the compilation of the resulting compute graph into efficient numerical code that operates on pre-allocated “column-centric” tensors, rather than the current “row-centric” format of dynamically allocated nested dictionaries.

This analysis led to an initial prototyping effort for a new runtime system that is compatible with the specification format of the Vivarium API, and that supports more general algebraic effects on top of a tensor-based memory management system, in turn following the *multilevel bulk synchronous* parallel model of computation (see SOP Sec. 8). While a functionally complete implementation of such a new runtime system is beyond the scope of this project, an immediate benefit of this effort is a new simulation emitter in the Xarray [24] format, which was developed for extracting training data for the new metabolism model from the tFBA process in the current WCM (see MS 5 Technical Report), and which is in preparation for upstreaming into the official Vivarium library.

High-performance stochastic process simulation. In light of the subsequent insight into PDMPs as a suitable model class for reformulating the WCM, it became clear that a significant fraction of the engineering effort described above would effectively be duplicating work that has already been invested in high-performance PDMP simulators [25]. This suggests two complementary avenues for future work in this line of investigation:

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- At the level of numerical model development — A staged research plan for constructing a new WCM in the PDMP class, in close collaboration with cell biologists and biochemical engineers, and starting from the nucleus of a metabolism model in the ODE class, as implemented in a library capable of high-performance PDMP simulation (see MS 6 Technical Report).
- At the level of software interface design — The development of a new version of the Vivarium API that enables the compilation of efficient PDMP problem representations [26] from modular specifications, in particular facilitating the incremental extraction and reformulation of existing WCM subsystems along with their ontologies. The primary architectural change in this refactoring is a kind of inversion of control, subordinating the combinatorial topology representation to an event-driven API that is general enough to support a broad range of composite numerical integrators. From the perspective of a user process, this amounts to a stricter decoupling of the local state space automaton from the global traversal structure that currently fulfills the roles of both topological coupling and time-stepping.

5 Conclusion

5.1 Original DUF ARC Goals

Within the restricted scope (Sec. 3), our project outcomes do not include causal discovery results, i.e., hypothesized function annotations with confidence scores or credible intervals. We provide updated resource estimates for future work in Sec. 5.3 below.

5.2 Key Outcomes and Next Steps

This project identified a theoretical, computational, and infrastructural foundation for a next generation of WCMs, which will be numerically suitable for a wide range of statistical and causal inference tasks, as exemplified by the DUF ARC goals. This new approach is centered around the following priorities:

- Incrementally transforming the current “mechanistic” WCM [1] (hybrid dynamical system heuristically combining a multitude of model types) into a mathematically well-defined **stochastic process model** (PDMP combining ODEs and jump processes). This type of numerical model will be easier to modify, communicate and validate, and more efficient to simulate.
- Augmenting existing experimental data pipelines with **measurement process annotations** that are required for establishing scientifically meaningful Bayesian likelihoods.
- Combining such a “dynamical system” WCM and observation likelihoods into a **Bayesian state space model** that is numerically much better behaved for the purposes of parameter inference and causal discovery.
- Increasing the available **independent sample sizes per experimental condition**, partly trading off against the dimensionality of omics data.

5.3 Lessons about Scientific Scope

Multidisciplinary expertise. Despite narrowing down the statistical problem scope early on, we repeatedly encountered biochemical modelling issues that could not be addressed, within the

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project timeline, to a degree that would allow for scientifically conclusive inference results. For future projects with comparable goals, we recommend assembling teams that combine all of the following specializations: *biochemistry* (e.g., pathway model development), *computational biology* (e.g., simulation infrastructure development), *applied mathematics* (e.g., numerical model development), and *computational statistics* (e.g., inference method development).

Computational experiment design vs. infrastructure development. Off-the-shelf algorithms for Bayesian parameter inference and causal discovery are not reliably scalable to the highly nonlinear, highly stochastic and high-dimensional structure of WCMs, calling instead for the development of bespoke inference methods, as is commonplace in other fields of scientific computing. In addition to a well-defined simulation-based statistical model, which was entirely lacking at the beginning of this project, this level of model complexity requires judicious choices about inference goals, simulation configurations and measurement data, which should be guided by concrete empirical questions and corresponding hypotheses about the relative relevance of WCM components. For future research, we thus recommend a methodological separation between projects addressing *concrete inference problems* in cell biology and projects developing *simulation-based inference infrastructure* for computational biology. While both types of projects need to plan for the multidisciplinary requirements above, and while they have to be well coordinated to advance jointly, they display different management priorities and risk profiles, and require somewhat different skill sets.

Resource estimates for future work. Based on the above discussion, it is important to distinguish between the two long-term goals that our DUF ARC proposal combined, namely gene function annotation and WCM development, in the pragmatic sense that prioritizing either of them alone would have led to fundamentally different project outcomes. For each of the goals classified in Sec. 3, we anticipate the following approximate resource requirements (see also MS 5 Technical Report):

1. Functional annotation based on non-mechanistic, static or stationary models can be treated within the classical paradigm of causal discovery in search spaces of static graphical models [8,9], and it is expected that many annotations can be either hypothesized or rejected successfully with such methods. This kind of approach is primarily a matter of appropriately combining model fragments, experimental measurements and existing algorithms, it is therefore suitable for teams and budgets comparable to the scope of the DUF ARC, and it would require much less computational resources than we had provisioned. In contrast, more complex approaches, using simulation models that can resolve dynamic, stochastic and nonlinear interactions, are called for when transient, heterogeneous or higher-order effects are suspected to play an important role, or when simpler methods make conflicting predictions.
2. Gray-box optimization approaches for WCM parameter calibration [1] may be considered similar in complexity to ensemble-based or Bayesian methods for parameter inference and causal discovery in “structured” or “physics-informed” machine learning models [7,10-12], each amounting to 1-2 year efforts for already qualified academic teams in cell biology and machine learning. The former approach relies heavily on context-specific expert knowledge, and has so far not achieved the type of numerical uncertainty quantification that is necessary for dynamical causal discovery. To date, the latter approach has only been computationally scaled to process models that can be considered simplified versions of small subsystems of WCMs, and has struggled to produce robust results for causal inference problems [27], which are strictly simpler than

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causal discovery problems. In fact, the broader conclusions drawn in [27] resonate with the ones drawn in this project.

3. The long-term research strategy we have identified is directly aimed towards the anticipated meeting point of the two dynamic approaches under (2), namely, Bayesian parameter inference and causal discovery [9,10-12] directly in WCMs, once the latter have been refactored in both numerical and software terms. This goal is topically much broader than the DUF ARC goal, but based on the analysis above, we hold that it constitutes the most reliable strategy for advancing WCMs to a state in which scientific problems like gene function annotation can be translated into standardized computational problems. Realising this vision requires parallel multi-year efforts by computational biologists, experimentalists, computational statisticians and software engineers (Sec. 5.2), but their coordination will be substantially simplified, compared to state of the art in WCMs [13], by the choice of PDMPs as the target model family. More specifically, with such a reformulated WCM already at hand, and assuming already well curated simulation protocols and empirical data sets, estimation of functional annotations with credible intervals translates into a challenging but well-studied class of Bayesian model comparison problems (involving intractable normalizing constants) for which dedicated approximation methods exist with *marginal likelihoods* [28], for which dedicated approximation algorithms can be developed in a 1-2 year effort led by a senior computational statistician. After the significant performance improvements enabled by the proposed simulator reformulation, the computational resources planned in our original proposal would be adequate for inference problems of this type; however, since for such complex models differences in inference algorithms can lead to orders of magnitude of differences in computational cost, the primary bottleneck should be understood as numerical expertise, rather than raw processing power.

5.4 Broader Recommendations for Future Research

Algorithmic standards for cross-model alignment. Despite extensive efforts to create standardized ontologies (e.g., BioCyc) and model formats (e.g., SBML), the alignment of models across publications in cell biology remains a major practical challenge. A lot could be gained from executable formal specifications, in the sense of model checking, for describing how newly published models relate to established ontologies. Beyond identifier mappings (see MS 3 Technical Report), this should involve specifications for *levels of description* of chemical species and non-elementary reactions, for *coverage properties* of pathway networks, and for *environment conditions* under which models were parametrized or simulated. Crucially, such specifications should be meaningful across model classes, e.g., across kinetic and constraint-based models of metabolism, and their emphasis should be on the *curation level of individual models*, rather than on heuristic aggregation of large numbers of models [29].

Mathematical standards for observation likelihoods. The measurement modalities dominating contemporary cell biology are much less varied than the molecular phenomena being studied. A library of observation likelihoods that are informed by the underlying physical processes of all common instrument types could contribute significantly to the development of statistical methodology, as has been the case in the field of GWAS.

5.5 Other Research and Funding Opportunities Enabled

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This ARC effort directly laid the foundation for my team's involvement in the DARPA Simulating Microbial Systems (SMS) program, which uses the same vivarium-ecoli model framework for applications in biomanufacturing, antibiotic resistance, and general simulation software. The SMS Proposer's Day aligned with the start of the DUF program, so the vision leading and developing this DUF project quickly manifested as a multidisciplinary team to extend this vision at a much larger scale, which was subsequently funded and is currently underway.

6 References

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